Circular Queue Implementation Using Array

C Program Code and Execution Output

PROGRAM SOURCE CODE

```
#include <stdio.h>
#define SIZE 5

int items[SIZE];
int front = -1, rear = -1;

int isFull()
{
    if ((front == rear + 1) || (front == 0 && rear == SIZE - 1))
        return 1;
    return 0;
}

int isEmpty()
{
    if (front == -1)
        return 1;
    return 0;
}

void enQueue(int element)
{
    if (isFull())
    {
        printf("\nQueue is full!!\n");
    }
    else
    {
        if (front == -1) {
            front = 0;
            rear = 0;
        } else {
            rear = (rear + 1) % SIZE;
        }
        items[rear] = element;
        printf("\nInserted -> %d", element);
    }
}

int deQueue()
{
    int element;
    if (isEmpty())
    {
        printf("\nQueue is empty!!\n");
        return -1;
    }
    else
    {
        element = items[front];
        if (front == rear)
        {
            front = -1;
```

PROGRAM OUTPUT

```
Queue is empty!!

Inserted -> 1
Inserted -> 2
Inserted -> 3
Inserted -> 4
Inserted -> 5
Queue is full!!

Front -> 0
Items -> 1 2 3 4 5
Rear -> 4

Deleted element -> 1

Front -> 1
Items -> 2 3 4 5
Rear -> 4

Inserted -> 7
Front -> 1
Items -> 2 3 4 5 7
Rear -> 0

Queue is full!!
```